

Door Subsystem — Engineered Features

Custom features we built for the Nebula door predictive-maintenance model • reference for `door_predict.features()`

What this is. The raw data is a 50 Hz stream of motor and sensor readings. We split it into individual door **open/close cycles** (segments), then turn each cycle into a fixed **31-number feature vector** that describes how that cycle behaved. These features are what the classifier sees — the raw waveform is never fed in directly. Every feature was designed around one physical idea: **abnormal resistance forces the motor to spend more electrical effort to move the door**.

Notation. For one cycle: I = motor current (mA), V = motor voltage (10 mV units), B = back-EMF ($\propto$ motor speed), P = door leaf position. n = number of samples in the cycle, $FS = 50$ Hz (so $dt = 1/FS = 0.02$ s). $\text{diff}(x)$ = sample-to-sample change, trapz = trapezoidal integral over time.

The 31 features fall into three families: **amplitude** (how much electrical effort), **resistance** (effort per unit motion — the core signal), and **frequency** (judder / intermittent binding).

1 Cycle basics (2 features)

Baseline descriptors of the cycle itself, used as context and for normalisation.

Feature	Formula	Why we made it
dur	n / FS	Cycle duration in seconds. A door that stalls or drags could run longer than a clean cycle.
n	$\text{len}(\text{cycle})$	Number of samples in the cycle. Raw length; several other features are normalised by motion rather than by this.

2 Amplitude family (7 statistics $\times$ 3 signals = 21 features)

Idea: the harder the door is to move, the more electrical effort the motor spends. We summarise each of the three electrical signals — current (I), voltage (V) and back-EMF (B) — with the same 7 statistics. In the table below X stands for any one of I , V , B , giving features named I_mean , V_mean , B_mean , I_max , ...

Feature	Formula	Why we made it
X_mean	$\text{mean}(X)$	Average level over the cycle — overall effort.
X_max	$\text{max}(X)$	Peak effort. A binding point shows up as a higher spike.
X_std	$\text{std}(X)$	Spread within the cycle — how uneven the effort was.
X_p90	$\text{percentile}(X, 90)$	High-end level, robust to a single freak sample (a gentler 'max').
X_auc	$\text{trapz}(X, dx=1/FS)$	Area under the signal over time — total accumulated effort, not just the peak.
X_rough	$\sqrt{\text{mean}(\text{diff}(X)^2)}$	RMS of sample-to-sample change — 'roughness'. Jitter/chatter raises it.
X_kurt	$\text{kurtosis}(X)$	Spikiness / heavy tails. Sharp isolated spikes (a catch) push kurtosis up.

$\Rightarrow$ computed for I (current), V (voltage) and B (back-EMF).

3 Resistance family (6 features) — the core signal

Idea: raw current alone is ambiguous (a longer or faster move also draws more). What actually indicates a fault is **current per unit of motion** — effort normalised by how much the door actually moved. This family measures resistance *directly* and, in our analysis, separates Normal from Abnormal the most cleanly.

Feature	Formula	Why we made it
pos_range	$\max(P) - \min(P)$	Total travel of the leaf in the cycle (sanity / normaliser).
pos_rate	$\text{pos_range} / n$	Travel per sample — a door-speed proxy.
work	$\text{trapz}(I \cdot V, dx=1/FS) / 1000$	Electrical work done by the motor over the cycle (energy). Rises when the motor fights resistance.
I_per_pos	$\text{mean}(I) / (\text{mean}(\text{diff}(P)) + 1e-6)$	Current per unit motion — resistance measured directly. High = lots of current for little movement.
I_per_bemf	$\text{mean}(I) / (\text{mean}(B) + 1e-6)$	Current per unit speed (back-EMF $\propto$ speed). A cleaner resistance ratio, unaffected by position noise.
stall_frac	$\text{mean}(\text{diff}(P) < 1)$	Fraction of the cycle the door barely moved — catches momentary stalls / sticking.

4 Frequency family (2 features)

Idea: some faults aren't about level but about *texture* — judder, chatter or intermittent binding. We take the power spectrum of the current signal (Welch's method, after removing its mean) and summarise where the energy sits.

Feature	Formula	Why we made it
cent	$\text{sum}(f \cdot \text{PSD}) / \text{sum}(\text{PSD})$	Spectral centroid — the 'centre of mass' of the current's frequency content. Judder shifts energy to higher frequencies, raising it.
hf	$\text{sum}(\text{PSD}[f \geq 10]) / \text{sum}(\text{PSD})$	Fraction of spectral power above 10 Hz — flags high-frequency chatter from intermittent binding.

PSD and f come from `welch(I - mean(I), fs=FS)`.

5 Which features actually mattered

On the 110 labelled cycles, the strongest separators (by Cohen's *d*, Abnormal – Normal) were all from the amplitude/resistance families — and this was confirmed independently by SHAP:

Feature	Family	Cohen's d	Reading
V_auc	amplitude	+3.63	more accumulated voltage effort
I_per_bemf	resistance	+3.60	more current per unit speed
I_per_pos	resistance	+3.42	more current per unit motion
I_auc	amplitude	+3.41	more accumulated current
pos_rate	resistance	-0.03	door speed — basically no difference
dur	basics	+0.03	cycle length — basically no difference

Takeaway. The door is **position-controlled**: when it meets extra resistance the controller pushes more current to still reach the same position in the same time. So the fault shows up as **electrical effort** (current/work ratios), *not* as a slower or longer cycle — which is exactly why our I_per_pos and I_per_bemf features work and why speed/duration features don't.

31 features total = 2 basics + 21 amplitude (7×3) + 6 resistance + 2 frequency. Source: `door_predict.features()`. Generated for the Nebula hackathon door subsystem.